

Final Test Report:

Homburg Appointment Booking System

1. Executive Summary

This report summarizes the validation results for the Homburg appointment booking project. The testing suite provides comprehensive coverage across functional, UI/UX, accessibility (BITV/WCAG), and negative scenario domains.

Total Test Coverage:

- **Automated Scenarios:** 18
- **Manual Test Cases:** 5
- **Audit Checklists:** 23 specific validation points (UI/UX & Accessibility)

Overall Status: Not Production Ready (Blocked by Critical & Blocker Defects).

Execution Environment: Chrome, Firefox, Safari (Desktop & Mobile).

2. Test Scope & Methodology

A. Automated Regression (Playwright)

The automated suite covers critical paths and system edge cases:

- **Core Flows:** Successful booking scenarios for "Bürgeramt" and "Fahrerlaubnis".
- **Logic & UI:** Service counter limits, filter interactivity, dynamic overview synchronization, and header/footer consistency.
- **Negative/Security:** Deep linking protection, session management (timeout/warning), form validation (input security) and API failure handling (500 error mocking).

B. Manual Auditing (Checklists)

The system was validated against high-standard quality requirements:

- **UI/UX & Responsive:** Layout integrity on 430px viewports, touch-target sizing, and modal behavior.
- **Accessibility (BITV 2.0 / WCAG 2.1):** Full keyboard navigation (Tab order, focus traps), screen reader semantics (ARIA labels, dynamic states), and contrast ratios.
- **Integration:** Validated external links, Google Maps navigation, and session isolation during modal interactions.

3. Execution Results

Category	Total	Passed	Failed / Blocked	Skipped / Deferred	Notes
Automated Scenarios	18	12	3	3	Failed: TS_08 (BUG_02, BUG_04), TS_13 (BUG_03). Skipped: TS_06, TS_11, TS_12
Manual Test Cases	5	4	1	0	Failed: TC_04 (BUG_01). Passed: TC_01, TC_02, TC_03, TC_05
Audit Checklists	23	23	0	0	23 verification points across UI/UX & Accessibility
TOTAL	46	39	4	3	100% Traceability maintained

4. Discovered Critical Issues (Bug Reports)

During testing, three defects were identified that impact data integrity and system stability.

ID	Title	Severity	Impact
BUG_01	Invalid special characters accepted in 'Jahr' field	Major	Allows invalid data entry, potentially causing database corruption
BUG_02	Inconsistent email validation across input fields	Critical	Prevents appointment confirmation delivery; renders service unusable
BUG_03	UI displays a blank page when Calendar API returns 500 error when transitioning to Step 4	Blocker	Disrupts the booking flow; user unable to select a date, leads to bounce
BUG_04	Invalid 4-digit postal code (PLZ) accepted as valid input on Step 5	Medium	Bypasses address validation, resulting in incomplete postal data

5. Accessibility & Responsive Compliance

The project underwent a rigorous audit following BITV 2.0 / WCAG 2.1 standards:

- **Responsive Integrity:** Confirmed 0 horizontal scrolling at 430px (Mobile).
- **Keyboard Accessibility:** Confirmed full navigability (Tab Order, Focus Traps in Modals, Escape Key functionality).
- **Semantics:** Verified ARIA states for key dynamic components (accordions, service counter buttons) to ensure proper screen reader feedback.

6. Conclusion & Recommendations

The system demonstrates good test coverage across core user flows, responsive design, and accessibility standards (BITV 2.0 / WCAG 2.1). Manual audits confirm full compliance with accessibility requirements, ensuring an inclusive user experience.

However, the application is currently **NOT PRODUCTION READY** due to active critical and blocker defects that prevent reliable deployment:

- **Data Integrity & Validation (BUG_01 - Major, BUG_02 - Critical, BUG_04 - Medium):** Input validation bypasses identified in the personal data step (Step 5) allow invalid year formats, incorrect email addresses, and invalid 4-digit postal codes (PLZ), preventing appointment confirmation delivery and risking database corruption.
- **System Resilience & UX (BUG_03 - Blocker):** The Calendar API failure when transitioning to Step 4 results in an unhandled blank screen, completely breaking the critical booking path without providing user feedback.

Recommendation: Resolution of BUG_02 and BUG_03 is strictly required as a prerequisite before any production release. Addressing these issues is vital to ensure service reliability and a seamless booking experience for all end-users.